CYPHER MODULAR CBIOS Version 3.1
SUMMARY SHEET

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The following is a summary of features and specs supported by the CP/M-68k modular CBIOS in C version 3.1 release 05-Jun-87 and later. The important features are character I/O interrupts and command line editing and recall capabilities, a fast bios resident enhanced terminal emulator (ADM3A, VT52, ANSI, TEK4010), and enhanced disk dma buffer space allocation and useage.

FILES

In addition to the standard CP/M-68k files necessary for system generation, the following are required:

- CB.SUB, LB.SUB - system generation command procedures
- CBROOT.C - root module
- CBCHARIO.C - character input/output module
- CBDISKIO.C - disk input/output module
- CBRSX.C - some resident system extensions module
- GRAPHI4.C - terminal emulator module
- CBASS.S - assembly language support module
- FONTTAB.S - character font table for terminal emulator

BIOS Functions

BIOS functions can be executed from assembly programs with the TRAP 3 command or from C programs with the BIOS((WORD)func,(LONG)D1,(LONG)D2) procedure in the CYPHLIB library. The following is an incomplete summary of features supported by the BIOS in addition to minimum system configuration features:

```
BIOS Function #
   17  - return aux device input status
   23  - return aux device output status
   24  - return console output status
   25  - special functions:
         D1.L = 0 - set abort enable flag to D2.L
                1 - set 4Hz interrupt counter register to D2.L
                2 - return 4Hz interrupt counter register in D0.L
                3 - set 4Hz interrupt enable flag to D2.L
                4 - set enhanced bdos enable flag to D2.L
                5 - set diagnostic mode enable flag to D2.L
                6 - set output spooler enable flag to D2.L (not implemented)
                7 - set cursor blink enable flag to D2.L
  version 3.2   8 - set baud rate on A to D2.L
                9 -  "    "    "   " B  "   "
```

BDOS Functions

BDOS functions can be executed from assembly programs with the TRAP 2 command for the enhanced server or TRAP 0 for the unenhanced server, or from C programs with the BDOS((WORD)func,(LONG)D1,(LONG)D2) procedure in the CYPHLIB library (the enhanced server). The following is a list of BDOS enhancements:

```
BDOS Function #
   10  - get command line     SUPPORTS COMMAND LINE EDITING AND RECALL
         ^W - recall previous line
         ^E - recall next line
         ^A - cursor left
         ^F - cursor right
         ^B - cursor to start or end of line
```

^G - delete character
^H - backspace and delete character
^R - repeat line
^U - cancel line (does not erase line)
^X - cancel line (erases line)
^C - warm boots if first character in line

TRAPS

The following is a summary of uses made of the 68000 software traps:

<u>Trap #</u>

Trap #	
0	- unenhanced BDOS server
1	- unused, reserved for system implementations
2	- enhanced BDOS server (normal BDOS entry)
3	- enhanced BIOS server (not normally used by programs)
4-7	- unused, recommended for program use
15	- terminal emulator

INTERRUPTS

The following is a list of 68000 interrupts implemented in the BIOS:

Interrupt #	
0	- 4 Hz interrupt (controls floppy motor auto stop and cursor blink)
1	- Character I/O (includes abort character check)
2	- Hardware abort and warm start (hardware not normally implemented)

DISK DRIVER

The floppy disk driver supports auto-density detection and track buffering. Not all density formats are implemented. See source code if unsure. The present version supports 8" SSSD, 8" SSDD, 8" DSDD and 5" 40 track DSDD. Track buffers are allocated dynamically during cold start. Permanent drive buffers are no longer released during a warm start. In addition there is a 5 Mbyte and a 20Mbyte hard disk driver and a Ram disk driver. The ram disk uses memory starting at $40000.

TERMINAL EMULATOR

A terminal emulator driver, upwards compatible with the EPROM resident emulator, is now included in the BIOS. All source code is supplied. The emulator supports ADM3A, VT52, TEK4010 and ANSI subset modes of emulation. See the source code for more documentation. The cursor is drawn only by an interrupt routine, not after each character, increasing the effective baud rate to 16600 baud if a 32MHz video clock is used. The emulator code should be entered with a TRAP 15 instruction. Enter with D0.W = 0 to reset, 1 for character output, and D1.W = character. Output will be redirected to the EPROM emulator if the BIOS emulator is not active.

Also: ^O is an interrupt with warm start when entered on the console device (abort).